

Population Distributions of Footbridge Modal Properties: A PRISMA-Guided Review and Hierarchical Bayesian Analysis

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Abstract

Keywords:

1. Introduction

The increasing adoption of light weight materials and slender structural forms has made modern footbridges more susceptible to human-induced vibrations. As a result, vibration serviceability under pedestrian loading has become an important design consideration to ensure acceptable structural performance and user comfort. Such assessment depends on several aspects of the bridge, including its structural form, material properties, and dynamic characteristics [1, 2, 3]. Therefore, prior knowledge or reliable estimation of key modal parameters is essential at the design stage, as these parameters provide an initial basis for assessing the feasibility of a proposed design [4, 5, 6, 7].

The current vibration serviceability codes and guidelines represent pedestrian loading through simplified load models to enable practical application at the design stage [8, 9, 10, 11, 12, 13, 14]. Since these load models are used to assess the vibration response of a wide range of footbridges, their adequacy should be assessed primarily in terms of the resulting structural response, rather than solely through their applied pedestrian loading.

The key limitation of many existing simplified load models is that pedestrians are represented as moving forces, with the corresponding load models calibrated without explicitly accounting for the dynamic interaction between the pedestrians and the supporting structure [15, 16, 17]. Human structure interactions (HSI) alters the dynamic properties of the coupled pedestrian structure system, and can lead to structural responses that differ substantially from those predicted by simplified load models. These effects can be

31 particularly important for lightweight structures and for cases with relatively
32 large pedestrian-to-structure mass ratios [18, 17]. Consequently, simplified
33 code models calibrated without accounting for HSI may exhibit systematic
34 response bias that varies with the dynamic characteristics of the bridge, mo-
35 tivating their reassessment and response-based recalibration.

36 Response-based calibration provides such a framework by comparing the
37 response predicted by the simplified design model with that obtained from ex-
38 perimental observations or a higher-fidelity reference model over a prescribed
39 range of structural properties. Previous studies have adopted this approach
40 by evaluating simplified pedestrian loading or vibration serviceability mod-
41 els over ranges of natural frequency, damping and other structural param-
42 eters [19, 3, 17, 18, 20]. However, these structural domains considered in
43 such studies have generally been defined using prescribed parameter ranges,
44 idealised structural configurations, or regularly sampled parameter combina-
45 tions rather than through the observed statistical distribution of dynamic
46 properties within the footbridge population. As a result, the contribution of
47 different structural conditions to the overall assessment is determined by the
48 adopted sampling scheme, rather than by their prevalence in practice.

49 **2. Systematic database development**

50 **3. Bayesian treatment of incomplete bridge data**

51 **4. Discussion**

52 **5. Conclusions**

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